

Mechanical sensors

Force-Amplifying Compliant Mechanism for Closed-Loop MEMS Accelerometer

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Abstract—Compliant mechanisms have been effectively utilized in microelectromechanical systems (MEMS) accelerometers to enhance the sensitivity of open-loop accelerometers. Traditionally, these mechanisms have been employed for displacement amplification. However, in this letter, a compliant mechanism is used to amplify the control force within the MEMS accelerometer. The structure incorporates a pair of compound lever-type mechanisms to amplify control forces acting in both forward and backward directions along the sensing axis. Closed-loop control is crucial for maintaining the linear characteristics of the MEMS accelerometer across a wide measurement range. Electrostatic force feedback is applied by providing dc voltage to the respective comb fingers for closed-loop implementation. As the input acceleration increases, the control voltage also rises. However, microlevel circuits have limitations on the maximum voltage that can be applied. The proposed force amplification strategy reduces the voltage required for position control. Consequently, the linear behavior of the MEMS accelerometer, the displacement of the proof mass, and the change in capacitance are maintained without the need for high voltage in electrostatic control.

Index Terms—Mechanical sensors, accelerometer, closed-loop, force-amplifying compliant mechanism (FaCM), microelectromechanical systems (MEMS), open-loop.

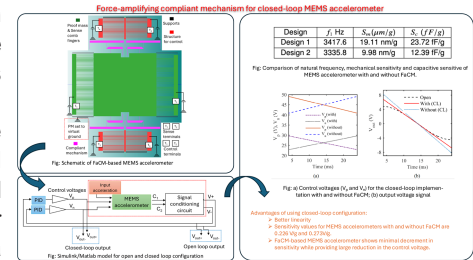
I. INTRODUCTION

Highly efficient capacitive microelectromechanical systems (MEMS) accelerometers are increasingly being utilized across a broad spectrum of motion-sensing applications due to their advantageous characteristics. These sensors, known for their compact size, low power consumption, and cost-effectiveness, have become essential components in various sectors including defense, automotive, industrial, and healthcare sectors for detecting acceleration, vibration, shock, tilt, and rotation [1], [2]. The demand for MEMS accelerometers extends to more advanced applications such as tactile sensing in robotics and navigation-grade sensing in aerospace, where the requirements for higher bandwidth and a broader measurement range are critical [3], [4]. To achieve a higher bandwidth, the accelerometer must be designed with either a higher natural frequency or increased stiffness, both of which typically result in reduced sensitivity. Therefore, a tradeoff must be made between sensitivity and natural frequency (or bandwidth) for performance improvements. To address this challenge, the concept of a compliant mechanism has been utilized. By incorporating a compliant mechanism, displacement amplification has been demonstrated at the microscale for capacitive MEMS accelerometers. This approach has been found to increase sensitivity without compromising natural frequency or bandwidth. [5], [6]. However, open-loop MEMS

accelerometers suffer from nonlinear output behavior. To accurately measure a wide range of accelerations, the sensor's output signal must remain linear within the desired sensing range.

As the range of input acceleration increases, the linearity of capacitive sensors tends to decrease. This is due to the inherent nonlinearity in the signal conditioning circuit and the nonlinear relationship between the change in capacitance and the displacement of the electrodes [7], [8]. By implementing the closed-loop control approach, the sensor's measurement range for input acceleration can be significantly increased. The closed-loop scheme effectively suppresses the displacement of the sense comb fingers, thereby reducing large changes in capacitance to achieve more linear sensor characteristics. As the input acceleration increases, a higher dc voltage must be applied within the closed-loop scheme to generate the necessary electrostatic control force. However, the maximum dc voltage that can be applied to MEMS devices is limited. Integrating compliant mechanisms within the closed-loop scheme can help reduce the required control voltage. At micro-nano scale, it is difficult to utilize large voltage values, as it affects the reliability of the device [9], [10]. Minimal power usage is another feature which can be achieved through the reduction of voltage for the feedback operation. Power efficient devices are always appreciated in the field of MEMS devices [11].

For the open-loop MEMS accelerometer, a compliant mechanism is used as a displacement-amplifying compliant mechanism (DaCM) to enhance the sensitivity. Different bridge and lever-type compliant mechanisms have been used for this purpose [5], [12]. In contrast, for the closed-loop approach, monolithic structures can function as a force-amplifying compliant mechanism (FaCM), which enhances the applied electrostatic control force. This letter presents the



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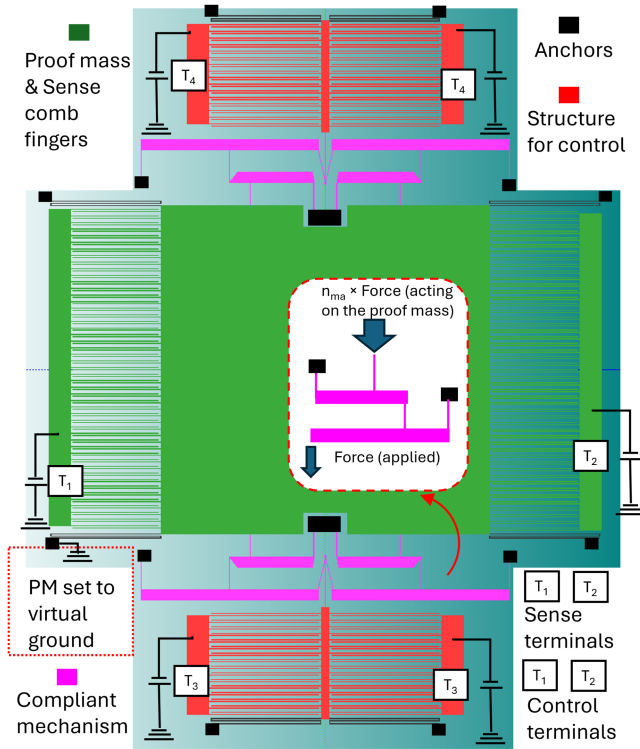


Fig. 1. Schematic of the MEMS accelerometer with FaCM included on both sides of proof mass.

implementation of this scheme and its outcomes. In this letter, a compound lever-type compliant mechanism is employed to amplify the electrostatic control force. The rest of this letter is organized as follows. Section II provides a detailed description of the FaCM-based MEMS accelerometer's structure, including the various electrodes used for capacitive sensing and actuation.

Therefore, this study examines both open-loop and closed-loop MEMS accelerometers, highlighting the use of a DaCM in open-loop systems and a FaCM in closed-loop systems. Section III details the closed-loop scheme and mathematical modeling, while Section IV compares the performance of MEMS accelerometers with and without FaCM. The findings indicate that the FaCM-based closed-loop MEMS accelerometer not only provides enhanced linearity but also operates effectively with reduced voltages for electrostatic force feedback, demonstrating a significant improvement over traditional models. Finally, Section V concludes this letter.

II. FORCE AMPLIFIER-BASED MEMS ACCELEROMETER

Capacitive detection is a well-established technology widely used in MEMS devices. This method is highly effective for generating output voltage signals in inertial sensing applications. Capacitive detection stands out due to its high resolution compared to other approaches. They also facilitate the straightforward integration of the mechanical structure with the signal conditioning circuitry, which enhances their practicality and efficiency in various applications [13].

Fig. 1 illustrates the operation of a closed-loop MEMS capacitive accelerometer, providing comprehensive details of its design. The principal components of the accelerometer are highlighted in different colors for clarity. The accelerometer features a proof mass supported

Table 1. Parameters of the MEMS Accelerometer

Parameters	Value
Dimensions of Proof mass	$1500 \mu\text{m} \times 1500 \mu\text{m} \times 100 \mu\text{m}$
Mass of proof mass	5.15 mg
Thickness for all the supports and comb fingers	7 μm
Layer thickness, b	100 μm
Comb fingers pair for sensing for either set	49 nos $\times$ 400 μm
Comb finger pairs for excitation for either set	32 nos $\times$ 500 μm
Sense gap	7 μm
Dimensions of larger and smaller levers	810 $\mu\text{m} \times$ 50 μm , 380 $\mu\text{m} \times$ 50 μm
Footprint area	2520 $\mu\text{m} \times$ 3216 μm

by folded beams. These beams act as structural supports that allow the proof mass to move in response to external forces. Movable comb-fingers are included to provide excitation, generating the necessary electrostatic forces for the sensor operation. The figure also illustrates the terminals dedicated to electrostatic sensing and control. These terminals are crucial for measuring the changes in capacitance, which are used to determine the acceleration. A key aspect of this design is the compound lever-type compliant mechanism, which is positioned both above and below the proof mass. This mechanism amplifies the applied force by a factor of 2.7. It includes two stages of amplification, which are designed to increase the mechanical advantage within the specified range. The designed dimensions and specifications of the accelerometer model are detailed in Table 1. Differential capacitive sensing is a technique used to measure changes in capacitance between two terminals, T_1 and T_2 , which are connected to the proof mass in a MEMS device. These terminals provide capacitance values C_1 and C_2 , respectively, reflecting the changes in the distance between the proof mass and the sensing electrodes as the mass moves. The proof mass is biased with a constant voltage V_m , which helps maintain a stable reference point for the capacitance measurements. To apply forces to the proof mass, two additional voltages are used: V_n and V_p . Voltage V_n is applied to the terminal T_4 , creating an electrostatic force that pushes the proof mass upward. Conversely, voltage V_p is applied to the terminal T_3 , generating an electrostatic force that pushes the proof mass downward. The differential measurement is based on the changes in capacitance C_1 and C_2 due to these forces. By comparing these capacitance values, the system can detect and quantify the movement of the proof mass. An additional feature of the system is the compliant force amplification mechanism, which is designed to amplify the electrostatic forces acting on the proof mass. This amplification improves the sensitivity of the accelerometer by increasing the force response to small changes in acceleration, thereby enhancing the overall performance and accuracy of the measurements.

III. CLOSED-LOOP SCHEME AND MATHEMATICAL MODELING FOR THE MODEL

Fig. 2 illustrates the components of both open-loop and closed-loop MEMS accelerometers, highlighting their respective structures and operational principles. In an open-loop MEMS accelerometer, the system generates an output voltage that is directly provided by the signal conditioning circuit. This voltage represents the accelerometer's measurement and is used to infer acceleration without any feedback mechanisms. Conversely, in a closed-loop MEMS accelerometer, the approach involves using a pair of output voltage signals. These signals are processed by a proportional-integral-derivative (PID) controller, which adjusts the voltages to provide electrostatic control. The PID controller compensates for any deviations and maintains the desired

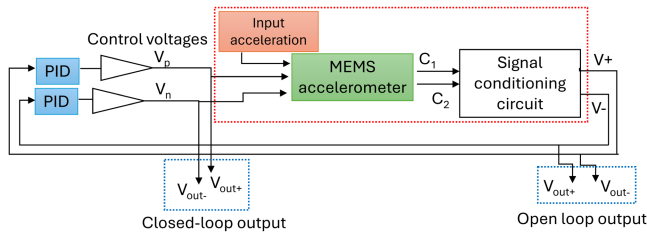


Fig. 2. Schematic of a closed-loop architecture of the MEMS accelerometer with FaCM.

Table 2. Parameters of PID Controller

Parameter	K_p	K_i	K_d
Value	21.64	0.001	0.001

performance by dynamically adjusting the control signals. The output of the closed-loop MEMS accelerometer is the difference between these adjusted control voltage values, reflecting the system's response to external forces with improved accuracy and stability. For a detailed explanation of the signal conditioning circuit used in these accelerometers, refer to the work of Tirupathi and Kar [13].

In the case of the closed-loop MEMS accelerometer, the voltages applied to the comb fingers mitigate nonlinear behavior, and this control action is used to generate the output voltage signal. The upward and downward control forces can be calculated as follows, incorporating the mechanical advantage (n_{ma}) as a multiplier for the electrostatic force:

$$F_{\text{up}} = n_{\text{ma}} \frac{\epsilon A_{\text{ex}} (V_n - V_m)^2}{2g_o^2}; F_{\text{down}} = n_{\text{ma}} \frac{\epsilon A_{\text{ex}} (V_p - V_m)^2}{2g_o^2} \quad (1)$$

The area of the parallel plate capacitor is given by $A_{ex} = n_{ec} b l_{ec}$. The voltages V_p and V_n can be expressed as $V_p = G(V_{cm} + V_1)$ and $V_n = G(V_{cm} - V_1)$, where V_{cm} is the common-mode voltage for the operational amplifier within the circuit, and V_m is the voltage applied to the proof mass structure. Both V_p and V_{cm} are considered to be 2.5 V. The gain G represents the overall gain applied to the voltage output values from the signal conditioning circuit, which includes the gain from the PID controller and the actuation gain. The term $2V_1$ denotes the differential voltage output generated by the signal conditioning circuit in response to the input acceleration (i.e., $V_0 = 2V_1$). This output is strongly dependent on the feedback capacitance of the differential operational amplifier (C_f) and the input pulse voltage (V_p) applied to the fixed electrodes for sensing (terminals T_1 and T_2) [13].

$$V_0 = V_+ - V_- = 8 \frac{\Delta C}{C_f} V_p \quad (2)$$

In this analysis, ΔC denotes the change in capacitance resulting from the applied input acceleration. The parameters n_{ec} and l_{ec} , which are crucial for calculating the capacitance, are detailed in Table 1. In addition, Table 2 provides comprehensive information on the PID gain values used in the system. For simplicity in this analysis, the actuation gain is considered to be unity, which means it does not alter the signal amplitude. The compliant mechanism integrated into the system provides a mechanical advantage, which has been calculated as $n_{ma} = 2.7$. This amplification factor enhances the force applied to the proof mass, improving the overall sensitivity and accuracy of the accelerometer. When the proof mass experiences very small or nearly suppressed displacements, the net electrostatic control force acting on the proof mass can be described mathematically as follows:

$$F_{\text{net}} = F_{\text{up}} - F_{\text{down}} \quad (3)$$

Table 3. Comparison of Different Performance Characteristics of MEMS Accelerometer in Open-Loop Configuration

Design	f_i Hz	$S_m(\mu m/g)$	S_c (fF/g)
Design 1	3417.6	19.11 nm/g	23.72 fF/g
Design 2	3335.8	9.98 nm/g	12.39 fF/g

$$F_{\text{net}} = n_{\text{ma}} \frac{\epsilon A_{\text{ex}}}{2g_0^2} (V_n - V_p)(V_n + V_p - 2V_m) \quad (4)$$

Equation (4) can be further simplified as taking $V_{cm} = V_m$ as follows:

$$F_{\text{net}} = -n_{\text{ma}} \frac{\epsilon A_{\text{ex}}}{2g_0^2} 4GV_1 V_m (G - 1). \quad (5)$$

It can be observed that the presence of the FaCM results in an amplification of the electrostatic control force by a factor of n_{ma} . This means that the FaCM effectively enhances the electrostatic force applied to the proof mass, with n_{ma} representing the mechanical advantage provided by the compliant mechanism. As a result, the overall sensitivity and responsiveness of the MEMS accelerometer have significantly improved due to this amplification effect.

IV. RESULTS AND DISCUSSION

The table presents data for two different designs of a MEMS accelerometer, showing their performance metrics in terms of natural frequency (f_1), mechanical sensitivity (S_m), and capacitive sensitivity (S_c). Design 1 (without FaCM) has a slightly higher natural frequency than Design 2 (with FaCM), suggesting that Design 1 might be stiffer or have a lower mass, making it less susceptible to lower frequency vibrations. Mechanical sensitivity refers to the displacement of the sensing element per unit of acceleration (in nanometers per g, where g is the acceleration due to gravity). Design 1 shows a significantly higher mechanical sensitivity, meaning it is more responsive to acceleration, producing a larger displacement for a given input force compared to Design 2. Capacitive sensitivity is a measure of the change in capacitance per unit of acceleration (in femtofarads per g). Again, Design 1 has a higher capacitive sensitivity, which indicates that it provides a stronger electrical signal (a larger change in capacitance) in response to acceleration compared to Design 2.

For the input acceleration signal in the range of ± 30 g, displacement of the proof mass and change in capacitance have been presented in Fig. 3. It shows that the implementation of FaCM causes higher suppression of the proof mass displacement to get linear behavior. Mechanical and capacitive sensitivity has been calculated using the reduced order model for the accelerometer, and the data has been shown in Table 3. The open loop sensitivity is lower for the FaCM-based model, as FaCM introduces more stiffness to the structure. Details of the natural frequencies have also been provided.

For the input acceleration signal shown in Fig. 3(a), control voltage values V_p and V_n have been presented in Fig. 4(a) for the MEMS accelerometer model with and without FaCM. One can identify that the required control voltages have been reduced by 40%. However, the linearity of the closed-loop MEMS accelerometer has not been affected by it. Fig. 4(b) presents the voltage output for the open loop approach and closed-loop approach for the MEMS accelerometer with and without FaCM. Sensitivity values for MEMS accelerometers with and without FaCM are 0.226 V/g and 0.273 V/g. FaCM-based MEMS accelerometer shows minimal decrement in sensitivity while providing large reduction in the control voltage.

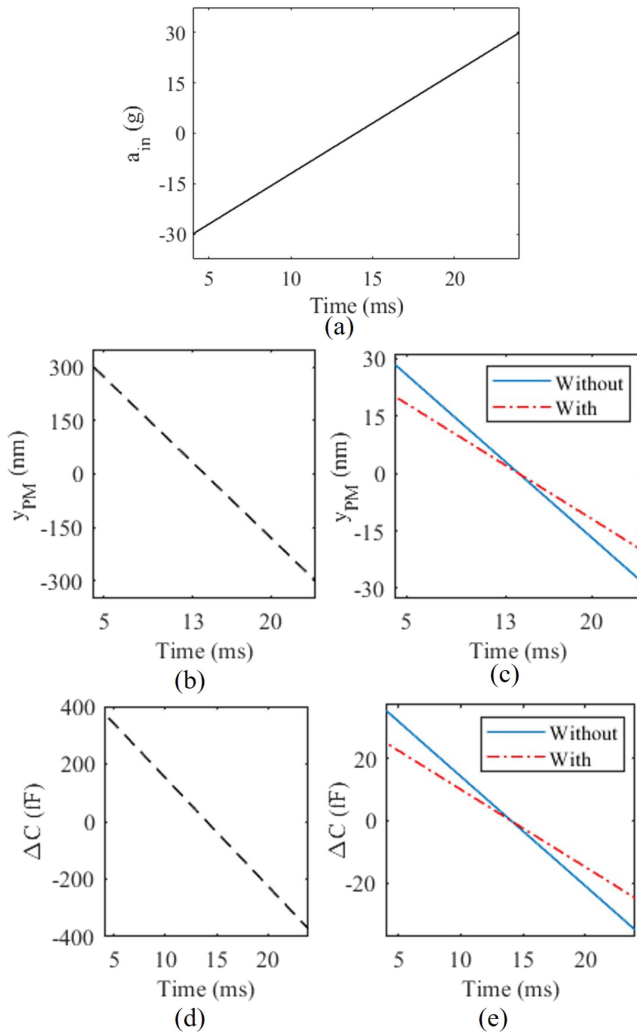


Fig. 3. (a) Input acceleration signal, change in capacitance for (b) open loop and (c) closed-loop with and without FaCM; PM displacement for (d) open loop and (e) closed-loop with and without FaCM.

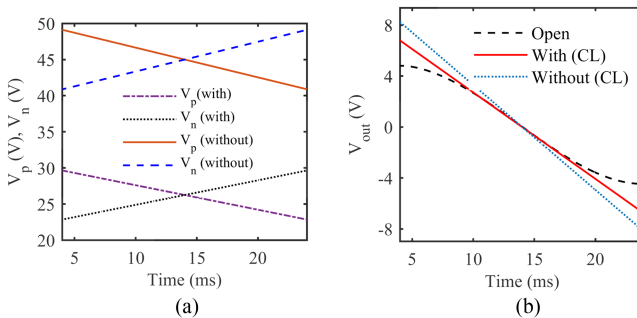


Fig. 4. (a) Control voltages (V_p and V_n) for the closed-loop implementation with and without FaCM. (b) Output voltage signal.

V. CONCLUSION

Open-loop MEMS accelerometers have benefited from the inclusion of compliant mechanisms to increase sensitivity through displacement

amplification. However, in the present research, the control force acting on the MEMS accelerometer has been amplified with the help of a compliant mechanism. For the MEMS accelerometer to exhibit linear characteristics over a wide measurement range, closed-loop control is necessary. To provide electrostatic forces, for controlling the motion in forward and backward directions, a separate set of comb fingers are provided, and a pair of compound lever mechanisms have been incorporated to amplify the upward and downward control forces. The proposed strategy of force amplification has demonstrated the reduction of voltage value for the closed-loop control. At the microscale, larger voltage values can not be applied due to limitations within a circuit. While utilizing, the FaCM, linear sensor output has been achieved within the range of ± 30 g input acceleration range with reduced values of applied voltages to the comb fingers. In the presence of input acceleration signals at higher frequencies, the dynamics of the lever of the compliant mechanism can play an important role. So, bandwidth analysis for the sensor can be carried out in the future, and closed-loop system parameters can be tuned for higher bandwidth.

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